Architecture and Design

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Introductions

Ground rules

Golden rule

Be respectful and mindful

Be timely about deliveries

Communicate

Cell phones in silence

Calls outside

AI is fine as long as you can explain it

Course Roadmap: Software Architecture & Design

Module	Weeks	Focus Theme	Key Learning Outcomes
1. Foundations	1 - 3	**The Architect's Mindset**	Identifying stakeholders, defining Quality Attributes (ASRs), and managing trade-offs.
2. Modeling	4 - 5	**Communication**	Mastering the **C4 Model**, Architectural Decision Records (ADRs), and visualization.
3. Component Design	6 - 8	**The Interior View**	Applying **SOLID**, Design Patterns (GoF), and refactoring for maintainability.
4. Macro Styles	9 - 11	**System Topology**	Choosing between Monoliths, Hexagonal/Clean Architecture, and Microservices.
5. Distributed Systems	12 - 14	**The Real World**	Handling Event-Driven design, CAP Theorem, Resilience (Circuit Breakers), and Cloud.
6. Synthesis	15	**Practice and summary**	Identifying Anti-patterns and preparing for Technical Leadership roles.

Summary of Activities and Exercises

Module	Core Activity / Exercise	Homework / Deliverable
1. Foundations	**The Stakeholder Negotiation:** Roleplay a conflict between a CEO and a Security Officer.	**Utility Tree:** Create a prioritized tree for a provided business case (e.g., a Fintech Startup).
2. Modeling	**C4 Mapping:** On-site modeling of a known system (like Netflix) on whiteboards.	**C4 Artifacts:** Submit a Level 1 (Context) and Level 2 (Container) diagram with 2 ADRs.
3. Component Design	**The SOLID Refactor:** Take a "God Class" and split it using design patterns.	**Pattern Implementation:** Code a small module using Observer or Strategy patterns.
4. Macro Styles	**Architecture Debate:** Defend "Monolith" vs "Microservices" for a specific budget.	**Style Analysis:** Written comparison of Hexagonal vs. Layered architecture for a new app.
5. Distributed Systems	**Chaos Workshop:** Simulate a server failure and design the "Circuit Breaker" logic.	**Resilience Design:** Diagram a saga pattern for a distributed payment transaction.
6. Synthesis	**Final Architectural Kata:** 60-minute "Mystery Prompt" design challenge.	**Self-Reflection:** Professional growth plan as a future Software Architect.

Architecture vs. Design (The Comparison)

Component	Weight	Description
Architectural Katas (x4)	**20%**	In-class design sprints. Graded on the logic of the diagrams and the justification of trade-offs (5% each).
Homework & ADRs	**20%**	Quality of documentation. Students must prove they can write professional **Architecture Decision Records**.
Midterm Assessment	**30%**	Evaluation of Modules 1, 2, and 3 (Foundations, Modeling, and Patterns).
Final Assessment	**30%**	Evaluation of Modules 4 and 5 (Macro and distributed patterns).
Total	**100%**	

Key aspects of grading

Architectural Katas: Fast-paced design challenges where they must solve a problem and justify it on the whiteboard.

Trade-off Defense: The ability to explain why you chose one path over another (The “It Depends” philosophy). No wrong answers.

Documentation Rigor: Moving beyond “boxes and arrows” to professional C4 diagrams and ADRs.